MIL-D-36776
July 16, 1971
(See 6.3)

# MILITARY SPECIFICATION

## DENTURE BASE FORMING KIT, FLUID RESIN TECHNIQUE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a kit and replacement items for the fabrication of denture bases by the use of the fluid resin technique of denture constructions.

1.2 Classification. The denture base forming kit and replacement items shall be furnished as follows (see 6.1):

1.2.1 Complete Kit. The complete denture base forming kit shall be furnished as specified in 3.2.

1.2.2 Replacement items. The replacement items shall be furnished in the following types, as specified:

| | |
|---|---|
| Type I. | Powder |
| Type II. | Liquid and Accelerator |
| Type III. | Denture flask |
| Type IV. | Plug Forming Instrument |
| Style A. | 3/16 inch |
| Style B. | 1/2 inch |

/FSC 6520/

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## 2. APPLICABLE DOCUMENTS

2.1 Specifications, standards, and drawing. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-A-225/8C | Aluminum Alloy Bar, Rod, Wire, and Special Shapes, Rolled, Drawn, or Cold Finished, 6061. |
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |
| PPP-B-676 | Box, Setup. |
| PPP-C-186 | Container, Packaging and Packing For Drugs, Chemicals and Pharmaceuticals. |

Military

| | |
|---|---|
| MIL-M-14 | Molding Plastics and Molded Plastic Parts, Thermosetting. |
| MIL-P-79 | Plastic Rod and Tube, Thermosetting, Laminated. |
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible. |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-66 | Steel, Chemical Composition and Hardenability. |



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STANDARDS (cont'd)

Federal

FED-STD-191 Textile Test Methods.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWINGS

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24196 Denture Base Forming Kit, Fluid Resin Technique.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified below, shall be of the highest quality, and shall be suitable in every respect for the intended purpose.

3.1.1 Types I and II resin denture material. Both the liquid monomer and powder polymer shall be essentially methyl-methacrylate resin with additives of a catalyst, a crosslinking agent, an ultraviolet absorber, fibers, and an accelerator.

3.1.1.1 Fibers. The fibers shall be fabricated of a polyacrylonitrile-type resin.

3.1.1.2 Type III, Denture flask. The denture flask shall be fabricated of materials specified in Table I for the components listed.



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Table I. Components and materials

| Quantity | Component | Material |
|---|---|---|
| 1 | Bottom plate | Aluminum alloy 6061 conforming to QQ-A-225. |
| 2 | Spring clips | AISI 301 or 302 conforming to FED-STD-66, annealed. |
| 1 | Inner sleeve | Laminated, cotton base, black phenolic tube conforming to Form Tr, Type FBE of MIL-P-79 except that color and electrical characteristics shall not apply. |
| 1 | Outer sleeve | |
| 1 | Top plate | Black phenolic molding material conforming to Type CFG of MIL-M-14 except that electrical characteristics shall not apply. |
| 1 | Center tube | |

3.1.1.3 Type IV, styles A and B. Plug forming instruments. The plug-forming instruments shall be fabricated of brass.

3.1.1.4 Measuring vials. The measuring vial for monomer shall be fabricated of clear glass, and the measuring vial for polymer shall be fabricated of cellulose acetate butyrate.

3.2 Description. The denture base forming kit shall be a group of ten completely compatible items which are supportive of the fluid-resin technique of denture base fabrication. The kit shall be supplied in a package containing a bottle of liquid, a bottle of accelerator, a bottle of powder, a denture flask, an assortment of three hydrocolloidal, plug-forming instruments, one set of two measuring vials for powder and liquid, and one set of complete instructions for use. A minimum of 35 heat-sealable, denture protecting bags shall be provided with each kit.

3.2.1 Liquid. When mixed, the liquid monomer and accelerator shall be compatible, and shall be clear, colorless, and substantially free from undissolved or particulate matter, when tested as specified in 4.4.1.1.



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3.2.1.1 Consistency after heating. The liquid shall not thicken when tested as specified in 4.4.1.2.

3.2.1.2 Discoloration. There shall be no visible discoloration of the exposed sample, when tested as specified in 4.4.1.3.

3.2.2 Accelerator. When warmed to 110 degrees Fahrenheit ( F.) in 120° to 130° F. water, the accelerator shall be liquid and shall be suitable for addition to monomer. The admixture shall be completely compatible with the polymer.

3.2.3 Powder. The polymer powder shall be pink in color and shall contain red fibers to provide characterization to the finished denture base. The fibers shall be 2 to 4 denier when tested in accordance with FED-STD-191, method 4021. The fiber length shall be 2.0 ± 0.2 millimeters. The powder shall be free from extraneous material such as dirt or lint.

3.2.4 Type III, Denture flask. The denture flask shall consist of seven parts, and shall be made up of a temperature-conductive metal bottom plate, a plastic inner sleeve, a plastic outer sleeve, a plastic top plate, a plastic center tube, and two metal spring clips. A stainless steel roll-pin shall be inserted in both the top and bottom plates for alignment with inner sleeve slots. The bottom plate shall have an anodized finish. The style, design, and dimensions of the flask shall conform to Defense Personnel Support Center (DPSC) Drawing 24196.

3.2.5 Type IV. Plug-forming instruments. The three plug-forming instruments shall be sprue cutters suitable for cutting smooth, precise holes for rapid and effective filling of the mold, and for venting, as required, in the hydrocolloidal duplicating material. The style, design, and dimensions of the instruments shall conform to DPSC Drawing 24196. Type IV, styles A and B plug forming instruments shall be available as replacement items.

3.2.6 Measuring vials. The measuring vial for monomer shall be suitably calibrated 1 through 12 cubic centimeters (cc.) with coordinated calibrations showing the recommended quantity for 1/4, 1/2, 3/4, and 1 denture unit. Measuring vial for polymer shall be graduated with manufacturer's recommended quantity for 1/4, 1/2, 3/4, and 1 denture unit. The one unit marking shall also be marked 30cc. The 30cc. of polymer and 12cc. of monomer for one denture unit represent the powder/liquid ratio (2.5cc. to 1.0cc.).



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3.3 Properties of uncured polymer. The uncured polymer, when used as directed by the instructions accompanying the package, shall be ready for pouring when the polymer is thoroughly wetted and dispersed in monomer after spatulating for 10 seconds. The mixture shall remain freely pourable for a minimum of 3 minutes from start of mix. When subjected to the surface roughness test, the mixed polymer shall be capable of showing a reproduction of the 32 microinch section of the roughness scale, when tested as specified in 4.4.3. Acrylic resin replacement, FSN 6520-871-7826, and acrylic resin liquid and catalyst replacement, FSN 6520-871-7827, shall be completely compatible with any batch or lot number of material and shall exhibit the properties enumerated in 3.2.1 and 3.2.3.

3.4 Properties of cured polymer. When processed according to the instructions furnished by the manufacturer, the polymer shall yield a satisfactory lightly veined denture with properties enumerated in 3.4.1 to 3.4.11.

3.4.1 Glossy surface and defects. When cured in a gypsum mold lined with tin foil, the polymer shall have a glossy surface and shall show no surface defects.

3.4.2 Polish. When polished by the conventional dental methods, the polymer shall present a smooth surface having a high gloss.

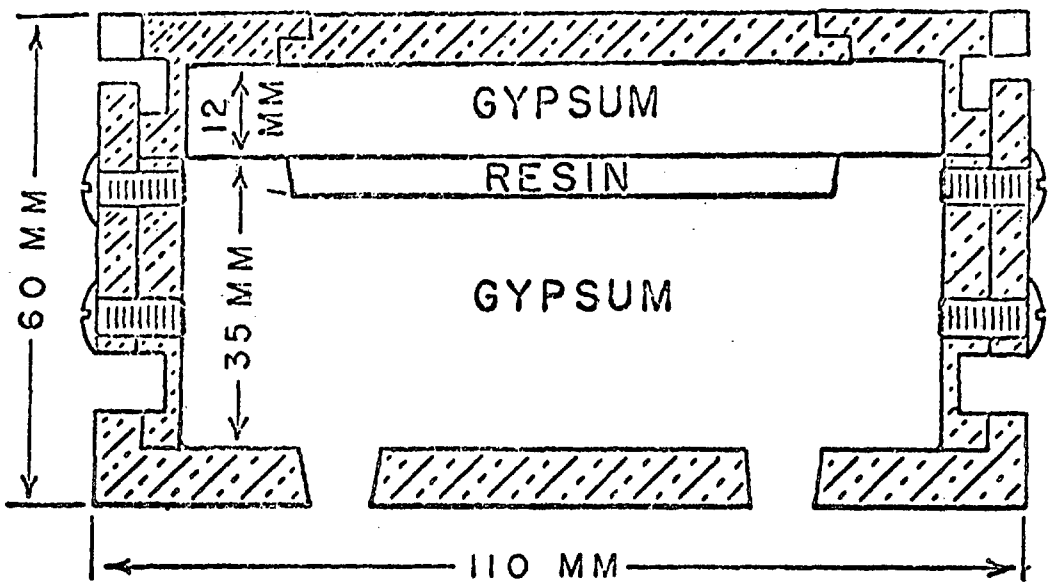
3.4.3 Toxicity. The manufacturer shall submit a statement that the denture base, produced according to the instructions supplied with the material, has no toxic effect on the normal, healthy person.

3.4.4 Color. The color of the pink polymer shall be equivalent to the standard sample (see 6.1). The color shall be uniformly distributed. A polished sample of the pink polymer shall be submitted by the manufacturer with the bid or proposal. The sample shall be a round disk 2-1/4 ± 1/8 inches in diameter by 0.125 ± 0.010 inch thick.

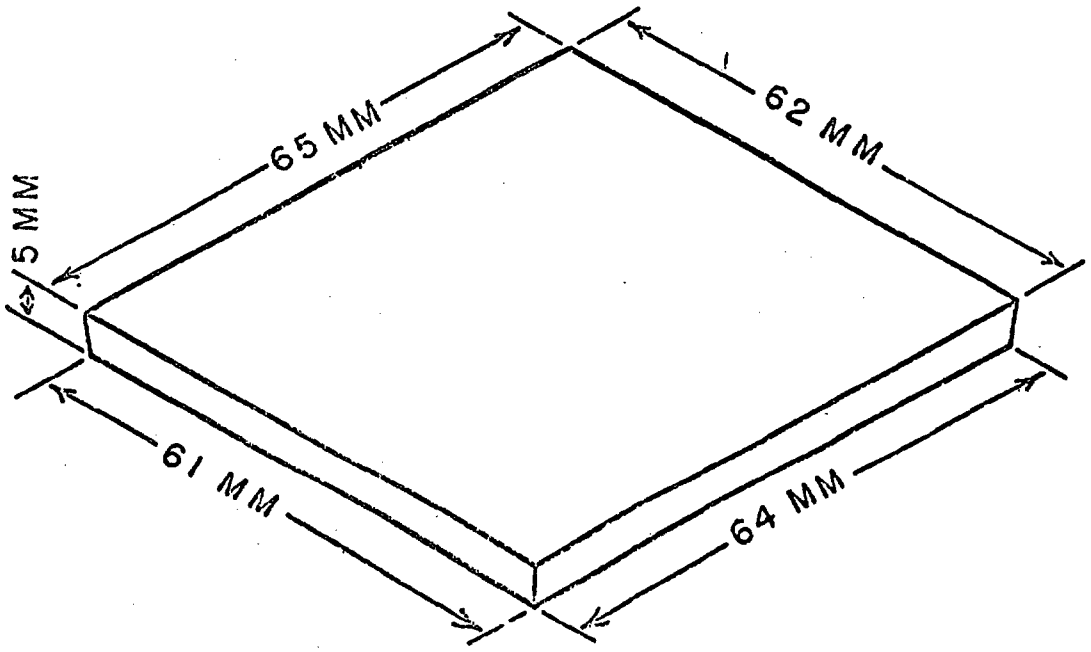
3.4.5 Translucency. The specimen plate of pink polymer, formed as described in 4.4.6, shall be translucent.



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DENTURE FLASK




SPECIMEN PLATE

**Figure 2**



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3.4.6 Porosity. The specimen plate (fig. 2), cut as described in 4.4.6, shall show no bubbles or voids.

3.4.7 Sorption. The increase in weight of the polymer shall be no more than 0.7 milligram per square centimeter (mg./sq. cm.), when tested as specified in 4.4.4.

3.4.8 Solubility. The loss in weight of the polymer shall be no more than 0.04 mg./sq. cm. of surface, when tested as specified in 4.4.5.

3.4.9 Transverse deflection. The deflection of at least three of the five specimens, prepared as outlined in 4.4.6, shall meet the requirements in table II.

Table II. Transverse Deflection Limits

| For 1500 gm. to 3500 gm. load:* | For 1500 gm. to 5000 gm. load:* | |
|---|---|---|
| Maximum deflection, mm. | Minimum deflection, mm. | Maximum deflection, mm. |
| 2.0 | 2.0 | 4.5 |

* Deflections shall be measured from an initial load of 1500 grams (gm.) to the load indicated.

3.4.10 Color stability. The polymer shall show no more than a slight change in color, when tested as specified in 4.4.7.

3.4.11 Hardening time, induction time, and peak temperature. The hardening time shall be a minimum of 14.0 minutes and a maximum of 24.0 minutes at a peak temperature of no less than 255° F. nor more than 285° F. The induction time (time from start of mix to a temperature of 110° F., when the rate of climb of the time-temperature curve has a definite change) shall be no less than 10 nor more than 15 minutes.



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3.5 Manufacturer's instructions. Instructions for storing the polymer, powder/liquid ratio (2.5cc. of powder to 1.0cc. of liquid), mixing and pouring times, and method and time of curing shall be prescribed by the manufacturer and shall accompany each unit of issue. In addition, the instructions shall include an illustrated description of the fluid-resin technique, and adequate and accurate instructions for using the polymer as a repair material. Two copies of all instructions shall be forwarded to the contracting officer for information purposes.

3.6 Shelf life. The monomer shall comply with the requirements of this specification for at least one year from date of shipment, when stored at 50° to 80° F.

3.7 Delivery schedule. No more than two months shall have elapsed from the date of manufacture of the resin denture material to the date of delivery to the Government.

3.8 Workmanship. Workmanship shall be first class throughout. The denture base forming kits shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.



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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the dental flask, plug forming instruments, and fiber, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identify has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the name of the manufacturer or supplier, the name of the component/material, the lot number, the date of testing, individual test results, and the requirements of this specification.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables III and V (see 4.2.1 and 4.3) and table IV (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table III. Unit of product for examination purposes shall be one denture base forming kit, or one replacement item, when applicable.

Table III. Acceptable Quality Levels

| Type defect | Inspection level | AQL (Percent defective) |
|---|---|---|
| Major A | II | 1.0 |
| Major B | S-1 | 1.5 |
| Minor A | II | 2.5 |
| Minor B | S-1 | 4.0 |



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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with Table IV. The acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one denture base forming kit.* Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Table IV Tests

| Characteristic | Requirement | Test Procedure | Number of determinations per lot or batch number |
|---|---|---|---|
| Clarity of liquid | 3.2.1 | 4.4.1.1 | 2 |
| Stability of liquid | 3.2.1.1 | 4.4.1.2 | 2 |
| Discoloration | 3.2.1.2 | 4.4.1.3 | 2 |
| Denier number | 3.2.3 | FED-STD-191, Method 4021 | 2 |
| Reproduction of detail | 3.3 | 4.4.2 | 2 |
| Glossy surface | 3.4.1 | /1 | 2 |
| Polish | 3.4.2 | /1 | 2 |
| Color | 3.4.4 | /1 | 2 |
| Translucency | 3.4.5 | 4.4.6 | 1 /2 |
| Porosity | 3.4.6 | 4.4.6 | 1 /2 |
| Sorption | 3.4.7 | 4.4.4 | 1 /3 |
| Solubility | 3.4.8 | 4.4.5 | 1 /3 |
| Transverse deflection | 3.4.9 | 4.4.6 | 1 /2 |
| Color stability | 3.4.10 | 4.4.7 | 1 |
| Hardening time and temperature rise | 3.4.11 | 4.4.8 | 2 /3 |
| Net content of powder and liquid | 5.1.1 | S. M. I. /4 | 2 |

*or one replacement item, when applicable.



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Table IV. Tests (cont'd)

/1 The polymer processed according to instructions furnished by manufacturer (3.4).

/2 Each determination shall be the average of 5 specimens.

/3 Each determination shall be the average of 2 specimens.

/4 Standard measuring instruments.

4.3 Examination. The denture base forming kit shall be examined for defects including, but not limited to, those listed in Table V.

Table V. Classification of Defects

| Categories | Defects |
|---|---|
| Major A: | |
| 101 | Unit not complete with powder, liquid, catalyst, denture flask, measuring vials, and plug forming instruments. |
| 102 | Powder not packaged in wide-mouthed glass bottles of suitable size. |
| 103 | Liquid not packaged in narrow-mouthed, light-resistant glass bottles of suitable design. |
| 104 | Accelerator not packaged in wide-mouthed, light-resistant glass bottles of suitable design. |
| 105 | Closure not of type specified. |
| 106 | Design of dental flask and plug forming instruments not as specified. |



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Table V. Classification of Defects (cont'd.)

| Categories | Defects |
|---|---|
| Major A: (continued) | |
| 107 | Manufacturer's instructions do not include:<br>a. Description, with illustrations of the fluid-resin technique.<br>b. Powder-liquid ratio.<br>c. Mixing and pouring times.<br>d. Method and time of curing.<br>e. Method for using the polymer as a repair material. |
| 108 | Components of dental flask do not properly assemble or disassemble. |
| 109 | Labeling not complete:<br>a. Bottles not commercially identified.<br>b. Bottles not marked with lot number.<br>c. Unit carton and intermediate carton not marked as specified. |
| 110 | Glass containers cracked or broken. |
| 111 | Liquid is turbid (not water clear). |
| 112 | Liquid not free from deposits or sediment (see 3.1.1). |
| 113 | Powder not free from extraneous material such as dirt and lint. |
| 114 | Instructions not included. |
| 115 | Cap does not have nonabsorbent lining material. |
| 116 | Liquid leaks from cap. |



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Table V. Classification of Defects (continued)

| Categories | Defects |
|---|---|
| Major B: | |
| 151 | Bottles in unit carton not secured against shifting or damage, or accidental opening of package. |
| 152 | Dimensions of denture flask affecting fit and function not as specified. |
| 153 | Fibers not of specified length. |
| Minor A: | |
| 201 | Labeling not legible or not free from defacing marks and scratches. |
| 202 | Lining material does not remain in cap when cap is removed. |
| 203 | Cap is not tight. |
| 204 | Unit package not as specified. |
| 205 | Components of dental flask not free from defacing marks and scratches. |
| 206 | Plug forming instruments do not cut smoothly. |
| 207 | Measuring vials not as specified. |



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Table V. Classification of Defects (continued)

| Categories | Defects |
|---|---|
| Minor B: | |
| 251 | Dimensions of denture flask (other than 152) not as specified. |
| 252 | Dimensions of plug forming instruments not as specified. |

4.4 Tests. Equipment and material shall be conditioned at 23.0° ± 2.0°C. for no less than 10 hours prior to testing.

4.4.1 Tests on liquid. Prior to testing, accelerator shall be added to liquid in accordance with manufacturer's instructions.

4.4.1.1 Clarity. The liquid shall be examined, without accessory magnification, against a black background and against a white background. Illumination shall be from a light which, at a point 10 inches from its source, provides an intensity of illumination of no less than 100 and no more than 350 foot candles.

4.4.1.2 Consistency after heating. The liquid shall be heated in the dark in a closed container at 60° ± 2° C. for 24 hours.

4.4.1.3 Discoloration. Discoloration of the liquid shall be determined by placing 10 milliliters (ml.) of the liquid into a clear glass test tube, having an inside diameter of approximately 1 cm., and by exposing this sample to bright diffuse sunlight for 24 hours. This sample shall be compared with 10 ml. of the unexposed liquid placed in a similar test tube. The tubes shall be viewed perpendicularly to their long axis when backed by a white background.

4.4.2 Preparation of specimens. The specimens shall be prepared in accordance with manufacturer's instructions furnished with the unit kit. Instructions are not applicable to the pouring and hardening tests (see 4.4.3 and 4.4.8).



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4.4.3 Pouring test. Previous to testing, the 32 microinch section of the roughness scale shall be boxed with wax or modeling clay. A resin sample shall be prepared according to manufacturer's instructions and, three minutes from start of mix, approximately 1/4 inch sample shall be poured into the boxed section. The sample shall be cured according to manufacturer's instructions. After curing, the cast specimen shall be separated from the roughness scale and both scale and specimen shall be placed under a binocular microscope on 15X magnification. The material shall be capable of showing a reproduction of the 32 microinch section of the roughness scale. The roughness scale shall be the General Electric Surface-Roughness Scale "Cat. No. 8665947G2," or approved equivalent.

4.4.4 Sorption. The water sorption test shall be made in duplicate on disks 50 ± 1 mm. in diameter and 0.5 ± 0.1 mm. thick. These specimens shall be made in a stainless steel die (fig. 1). The die shall be mounted with gypsum in the flask in the same relative position as the specimen plate in figure 2. The gypsum in the shallow half of the flask shall be poured against a glass plate to obtain a smooth surface. Before final closure of the flask, a sheet of tinfoil shall be substituted for the cellophane. All surfaces of the disk shall be smooth, and the top and bottom shall be flat. The disks shall be dried in a desiccator containing anhydrous calcium sulfate ($CaSO_4$) at 37° ± 2°C. for 24 hours, removed to a similar desiccator at room temperature for 1 hour, when weighed with a precision of 0.2 mg. This cycle shall be repeated until constant weight is attained, that is, until the weight loss of each disk is no more than 0.5 mg. in any 24-hour period. The disks shall then be immersed in distilled water at 37.0° ± 1.0°C. for 24 hours. Then the disks shall be removed from the water with tweezers, wiped for 30 seconds with a clean, dry hand towel, waved in the air for 15 seconds, and weighed 1 minute after removal from water. The value for water sorption shall be calculated as follows for each disk:

$$\frac{\text{Weight after immersion - conditioned weight (mg.)}}{\text{surface area (sq. cm.)}} = \text{sorption (mg. per sq. cm.)}$$



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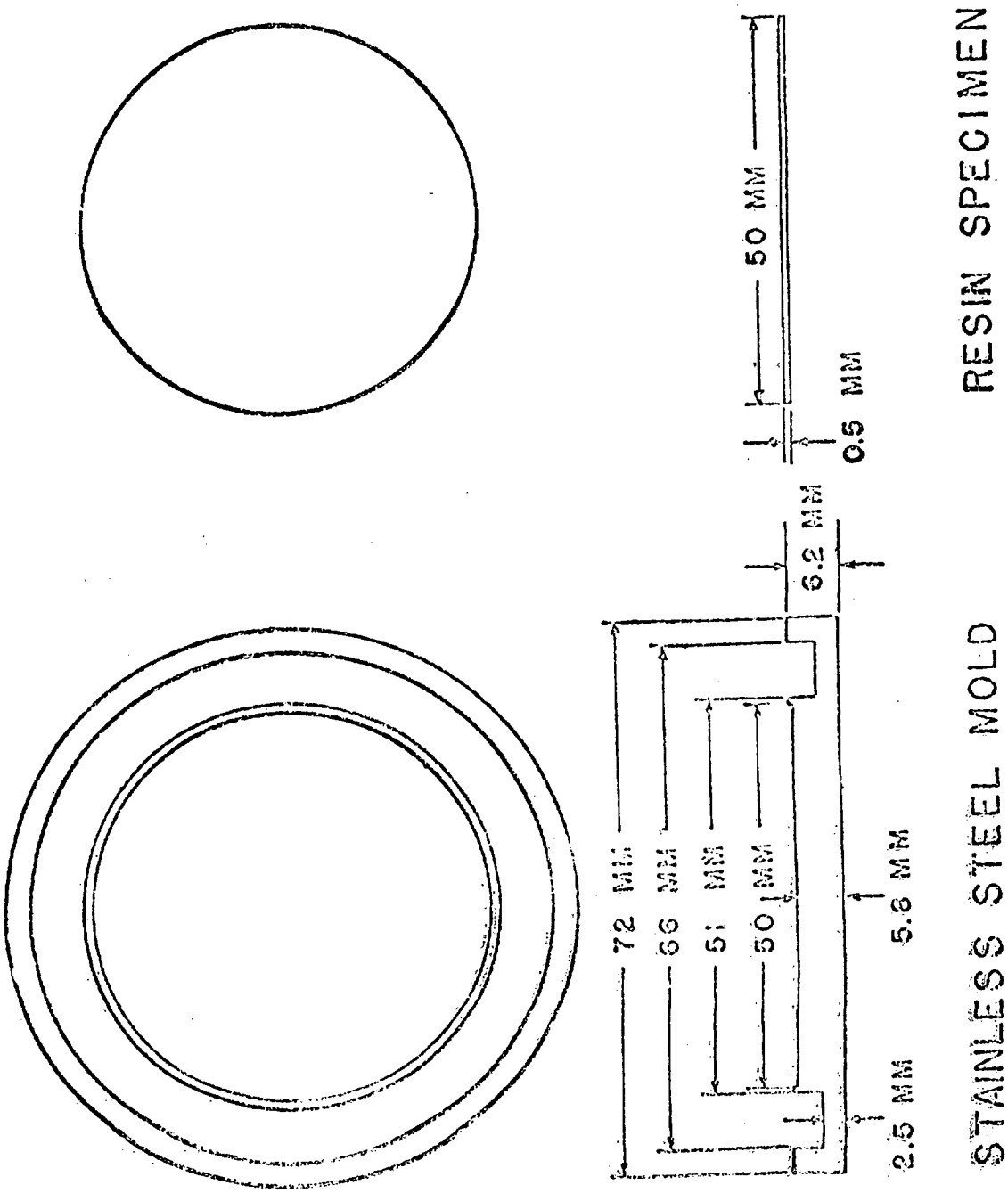



**Figure 1**



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The average of the determined values for the two disks shall be recorded to the nearest 0.01 mg. per square centimeter. The final value (average of two determinations) shall be rounded to the nearest 0.1 mg. per square centimeter. When the final value falls midway between the two numbers, the even number shall be recorded.

4.4.5 Solubility. After the final weighing (4.4.4) the disks shall be reconditioned to constant weight in the desiccator at 37° ± 2° C., as was done in 4.4.4. The soluble matter lost during immersion shall be determined as follows, to the nearest 0.01 mg. per square centimeter, for each disk:

$$\frac{\text{conditioned weight (mg.)} - \text{reconditioned weight (mg.)}}{\text{surface area (sq. cm.)}} = \text{solubility (mg. per sq. cm.)}$$

The final value (average of two determinations) shall be rounded to the nearest 0.01 mg. per square centimeter. When the final value falls midway between two numbers, the even number shall be recorded.

4.4.6 Transverse deflection tests. A plate of polymer shall be processed in a gypsum mold lined with tinfoil (fig. 3). Specimens for the transverse test shall be machined from this plate to the dimensions shown in figure 4. The plate (fig. 2) shall be sawed lengthwise into five equal strips with a band saw operated at a speed of approximately 100 meters per minute (328 feet per minute). For Type II polymers the plate shall be at least 2 days old prior to sawing. The strips may then be sanded slightly on the molded surfaces, if necessary, to remove imperfections. All subsequent milling operations shall be with a 50 mm. diameter flycutter, operating at a spindle speed of approximately 1,800 revolutions per minute and at a table feed of about 6 inches per minute. Rough cuts shall be approximately 0.08 mm. deep, and finish cuts shall be 0.03 mm. inch deep. The strips shall be mounted side by side in a vise, and the edges machined parallel in order to remove all band saw marks. The strips at this point shall be oversized in width. One or more strips shall be mounted in a vise and the faces milled to the required thickness of 2.50 ± 0.03 mm. All five strips shall then be clamped face to face in a vise and the edges milled to the required width of 10.00 ± 0.03 mm.



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The specimens shall be stored in distilled water at 37.0° ∱1.0° C. for 2 days immediately before testing. The specimen shall be mounted in an appropriate instrument such as shown in figure 3. An initial load, consisting of the free-moving parts of the apparatus (fig. 3) plus sufficient lead shot to make a load of 1, 500 gm., shall be placed on the specimen. Each 500 gm. increment of shot shall be added at a uniform rate during the last 30 seconds of each minute. The schedule shown in table VI shall be used in applying the loads and in making the observations.

Table VI. Schedule of loading and reading

| Time | | Load | Dial reading* |
|---|---|---|---|
| min. | sec. | gm. | mm. |
| 0 | 00 | 1, 500 | |
| 0 | 30 | 1, 500 | A |
| 1 | 00 | 2, 000 | |
| 1 | 30 | 2, 000 | |
| 2 | 00 | 2, 500 | |
| 2 | 30 | 2, 500 | |
| 3 | 00 | 3, 000 | |
| 3 | 30 | 3, 000 | A |
| 4 | 00 | 3, 500 | |
| 4 | 30 | 3, 500 | B |
| 5 | 00 | 4, 000 | |
| 5 | 30 | 4, 000 | |
| 6 | 00 | 4, 500 | |
| 6 | 30 | 4, 500 | |
| 7 | 00 | 5, 000 | |
| 7 | 30 | 5, 000 | C |

Deflection at 3, 500 gm. = B-A
Deflection at 5, 000 gm. = C-A

* Readings shall be made to the nearest 0. 01 mm.



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TRANSVERSE TESTING MACHINE

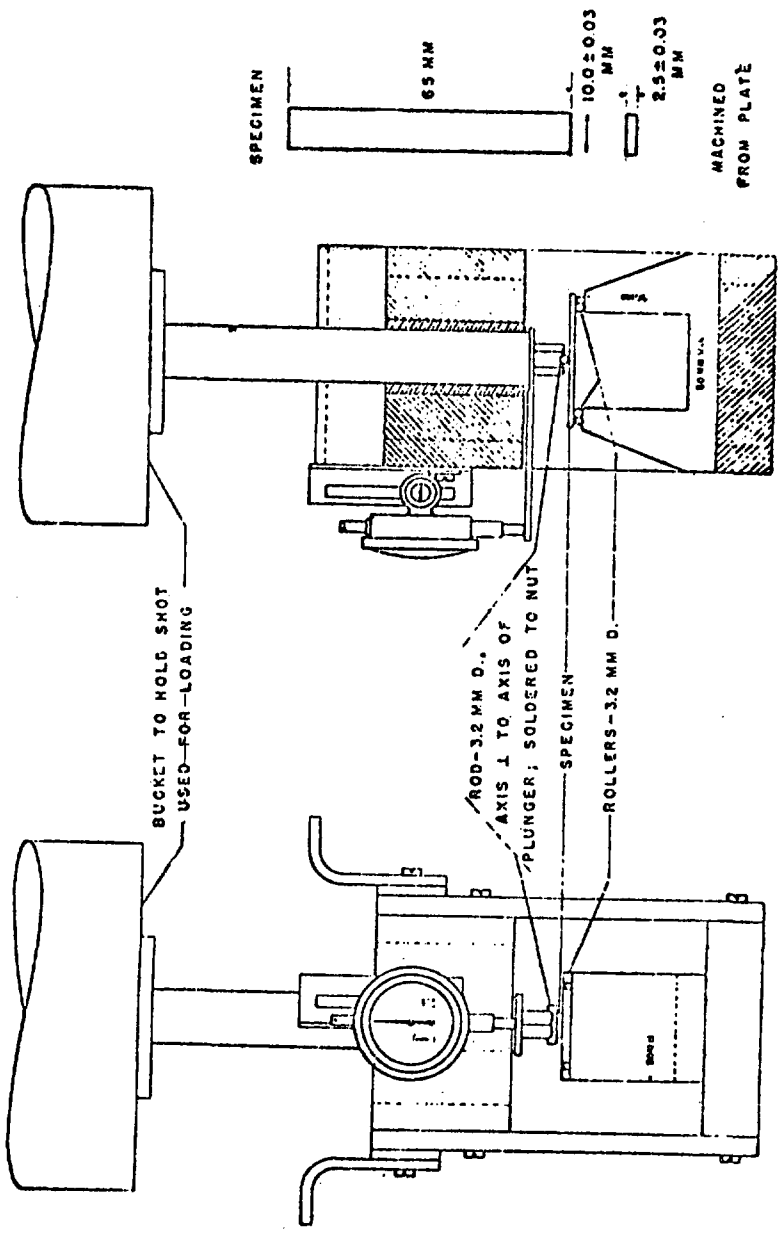



**Figure 3**



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The differences between the deflections at the initial load and at specified loads shall be taken as the deflections of the specimen. The recorded deflections shall be reported to the nearest 0.1 mm. When the determined number falls midway between two numbers, the even number shall be recorded.

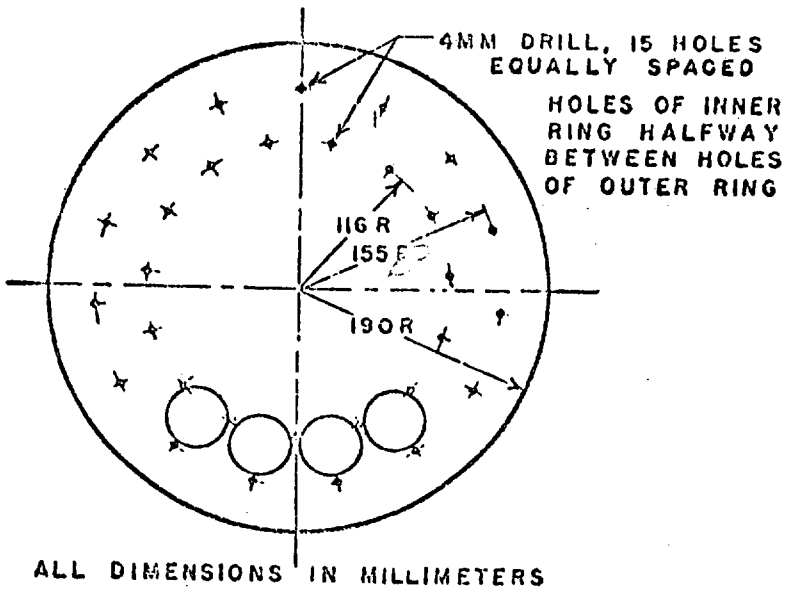
4.4.7 Color stability. One of the disks used in 4.4.4 and 4.4.5 shall be exposed to the radiation of a lamp (fig. 4) with an S-1 bulb which has been in use no less than 50 nor more than 400 hours. The light source shall be a combination tungsten filament mercury arc enclosed in Corex D or other glass which has a low transmission below 2,000 Angstrom units. The lamp shall be rated at 400 watts. The specimen shall be carried on an aluminum disk, which shall be centered under the S-1 bulb, and which shall rest on a phonograph turntable operated at 33 revolutions per minute. The specimen shall be held by screws approximately 0.2 in. above the disk and 5 in. from the center and shall be maintained between 60° and 65° C. The plane of the top surface of the specimen shall be 7 in. from the bottom of the S-1 bulb. After exposure for 24 hours, the specimen, compared with an unexposed duplicate specimen, shall show no more than a slight change in color (perceptible with difficulty). The comparison shall be made by visual inspection in daylight.

4.4.8 Hardening time. Twenty-five cc. of polymer powder shall be mixed with 10 cc. of liquid and shall be spatulated for 20 seconds. After a 30-second rest, the slurry shall be spatulated again for 10 seconds, and shall be immediately poured into a one ounce, waxed paper, medicine cup graduated in drams. An iron-constantan thermocouple shall be wound around a support bar in such a manner that the tip of the couple shall be centered in the cup at the 3 ± 1/2 dram mark when the bar rests across the top of the cup. Potentiometer readings shall be made at 30-second intervals beginning 1-1/2 minutes from start of mix and terminating two minutes after the maximum temperature has been reached. The time from start of mix to the time at which the peak temperature occurs shall be reported to the nearest 1/2-minute. In addition, the time from start of mix to time when induction temperature (110° E) occurs shall be reported to the nearest 1/2-minute. The times shall be the average values for two specimens. The values for each specimen shall agree within ± 1 minute.



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LAMP TO TEST FOR COLOR STABILITY



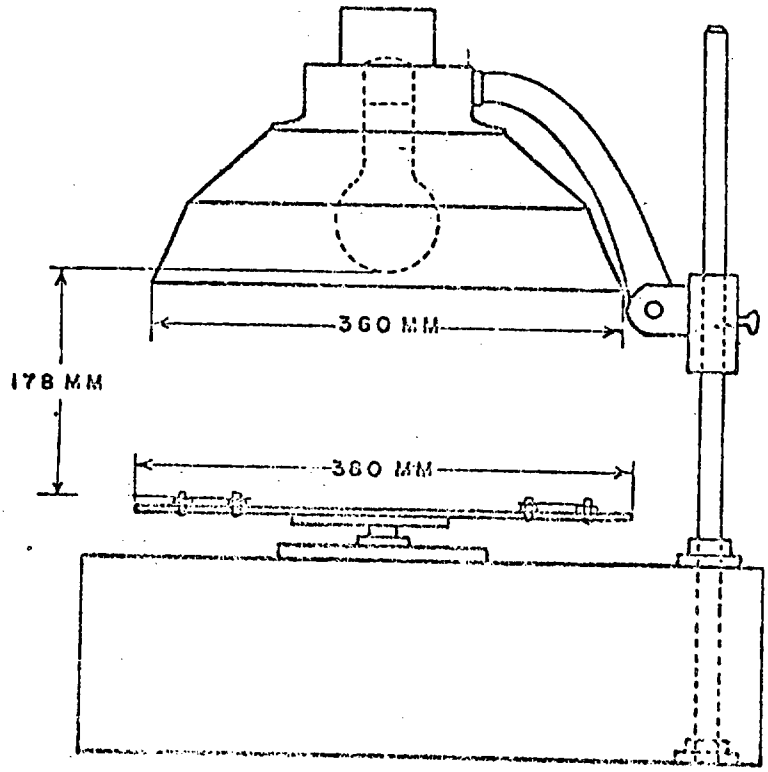




**Figure 4**



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## 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One each kit as specified in 1.2.1, or one bottle containing 650 grams minimum of powder (Type I) or one package, containing one pint of liquid and one bottle of accelerator (Type II) or one each denture flask (Type III) or one each plug forming instrument (Type IV, Style A or B) as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table VII. Packaging Quantities

| ITEM | Unit package | Intermediate package | Exterior container |
|---|---|---|---|
| KIT | 1 unit | Not Required | 4 units |
| Type I | 1 unit | Not Required | 12 units |
| Type II | 1 unit | 12 units | 24 units |
| Type III | 1 unit | 12 units | 24 units |
| Type IV | 1 unit | 12 units | 600 units |

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.



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5.1.3 Level A.

5.1.3.1 Unit package.

5.1.3.1.1 Type I. (Powder). Each unit shall be packaged in a bottle conforming to Group A, class 1, Type d, Style 2, Grade 2, Closure B or F, and Seal A, as specified in PPP-C-186. Style 2 shall have an inside diameter of no less than 1-3/4 inch. Bottle shall be further packaged in a carton of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.3.1.2 Type II (Liquid and Accelerator). Liquid shall be packaged in a bottle conforming to Group A, Class 1, Type c, Style 1, Grade 1, and Closure B or F, as specified in PPP-C-186. Style 1 shall have an inside diameter of no less than 0.528 inch. Accelerator shall be packaged in a bottle conforming to Group A, Class 1, Type e, Style 1, Grade 1 and Closure F, as specified in PPP-C-186. The bottle shall be further packaged in a carton of appropriate size and design. The bottle of liquid and the packaged accelerator shall be packaged together in a telescopic, spiral-wound, chipboard container of appropriate size. Closure shall be adequate to prevent accidental opening.

5.1.3.1.3 Type III (Denture Flask). Each unit shall be packaged in a carton of appropriate size and design constructed in accordance with PPP-B-566, or PPP-B-676. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent accidental opening.

5.1.3.1.4 Type IV (Plug Forming Instrument). Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material and packaged in a carton of appropriate size and design. Carton shall be constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.3.1.5 Kit. Components, comprising one unit of issue (one kit), packaged as specified in 5.1.3.1.1 thru 5.1.3.1.4 along with measuring vials, thirty-five heat-sealable bags and a complete set of instructions for use,



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shall be further packaged in a carton of appropriate size and design, constructed in accordance with PPP-B-636, type CF, class domestic. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure shall be adequate to prevent accidental opening.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.



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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Immediate containers. Immediate containers such as bottles and cartons, when not the unit package, shall be commercially identified.

5.3.2 Unit and Intermediate packages. Unit and Intermediate packages shall be marked in accordance with MIL-STD-129. Instructions for use shall be furnished with each unit package. For Kit, Type I and Type II items, marking shall include the lot (control) number and the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59 - 86 F.)."

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. For Kit, Type I and Type II items, marking shall include the lot (control) number and the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Complete kit or type and style of replacement item required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).



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6.2 Supersession data. This specification includes the requirements of the following Defense Medical Purchase Descriptions:

| P. D. | FSN | Item Identification |
|---|---|---|
| P. D. No. 1, dated 21 August 1969 and Modification No. 1, dated 22 February 1971 | 6520-125-9919 | DENTURE BASE FORMING KIT, Fluid Resin Technique |
| P. D. No. 1, dated 15 August 1968 | 6520-935-9870 | PLUG FORMING INSTRUMENT, DENTAL, Corrosion-Resisting Metal, 3/16 inch Diameter |
| P. D. No. 1, dated 16 August 1968 | 6520-935-9871 | PLUG FORMING INSTRUMENT, DENTAL, Corrosion-Resisting Metal, 1/2 inch Diameter. |
| P. D. No. 1, dated 13 August 1968 | 6520-871-7823 | FLASK, DENTURE, Upper and Lower: Seven-piece. |
| P. D. No. 1, dated 22 August 1968 | 6520-871-7826 | RESIN, ACRYLIC, DENTAL, Denture Base Material, Pink, 1-1/2 lb: Powder |
| P. D. No. 2, dated 21 November 1968 | 6520-871-7827 | ACRYLIC RESIN LIQUID, DENTAL, 1 pt. |

6.3 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.



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Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1290

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION

MIL-D-36776

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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